

OCR Further Mathematics
Pure Core - Polar Coordinates
Mark Scheme
AS and A Level
Further Mathematics A - H235, H245



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The latest topic questions for the year 2024 / 2025+ exams.
To be used by all students preparing for OCR AS and A Level Further Mathematics.
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Question			Answer/Indicative content	Marks	Part marks and guidance	
1	a		$\left(r = 0 \Rightarrow \sin 3\theta = 0\right) \Rightarrow \theta = 0, \frac{\pi}{3}$ $\Rightarrow 3\theta = 0, \pi$	B1 (AO 1.1) [1]	Both required Don't give if any extras within range. Ignore values outside range	
	b		$\left[\sin \frac{3\pi}{6}, \frac{\pi}{6}\right]$ i.e. $\left[1, \frac{\pi}{6}\right]$	B1 (AO 1.1) B1 (AO 1.1) [2]	For r For θ	
	c		DR $\text{Area} = \frac{1}{2} \int_0^{\pi/3} r^2 d\theta = \frac{1}{2} \int_0^{\pi/3} \sin^2 3\theta d\theta$ $= \frac{1}{4} \int_0^{\pi/3} (1 - \cos 6\theta) d\theta$ $= \frac{1}{4} \left[\theta - \frac{1}{6} \sin 6\theta \right]_0^{\pi/3}$ $= \frac{1}{4} \left(\frac{\pi}{3} - 0 \right) = \frac{1}{12} \pi$	M1 (AO 1.1) M1* (AO 3.1a) DepM1 (AO 1.1) A1 (AO 1.1) [4]	Correct use of formula - ignore limits Attempt to use double angle formula (Could be wrong way round, 2 missing or sign wrong) Integrate <i>their integrand</i> Use correct limits, must be seen	
	d		$\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$ $\Rightarrow r = \frac{3y}{r} - 4 \left(\frac{y}{r} \right)^3$ $\Rightarrow r^4 = 3r^2 y - 4y^3$ $(x^2 + y^2)^2 = 3y(x^2 + y^2) - 4y^3$ oe e.g. $(x^2 + y^2)^2 = 3x^2 y - y^3$	M1 (AO 1.1) A1 (AO 1.1) [2]	Using triple angle formula and $y = r \sin \theta$ isw	



Question			Answer/Indicative content	Marks	Part marks and guidance
			Total	9	



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Question		Answer/Indicative content	Marks	Part marks and guidance		
2	a	DR $\frac{1}{2} \int \left(\sqrt{\sin \theta} e^{\frac{1}{3} \cos \theta} \right)^2 d\theta$ $A = \frac{1}{2} \int_0^{\pi} \sin \theta e^{\frac{2}{3} \cos \theta} d\theta$ $= \frac{1}{2} \times -\frac{3}{2} \left[e^{\frac{2}{3} \cos \theta} \right]_0^{\pi}$ $\frac{3}{4} \left(e^{\frac{2}{3}} - e^{-\frac{2}{3}} \right)$	M1(AO3.1a) *A1(AO2.1) de p*M1(AO1.1a) A1(AO1.1)[4]	Correct form, in terms of θ, Integrand has been squared out. Must include limits (can be seen later) Might be as result of substitution Allow coefficient error for M1 isw	M1 can be implied by 1.0757... BC eg $\frac{3}{4} \left[e^{\frac{2}{3}} \right]_{\frac{1}{2}}^{\frac{3}{4}} \text{ or } \frac{3}{4} \left[e^{\frac{2}{3}} \right]_{-1}^1$ oe	

Examiner's Comments

Where an integrand is the product of two functions, there are three techniques available to candidates – inspection, substitution and parts. Candidates at this level should be encouraged to develop their ‘inspection’ skills sufficiently to recognise this integrand to within a constant term. The purpose of the other two techniques is to produce an integrand that has become simpler to integrate. Where candidates selected ‘parts’ (which a lot did) they produced a more complicated integrand; at this point they should reconsider their approach. Flexibility of approach to problem-solving is an important skill to develop.

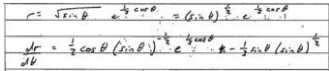


Question		Answer/Indicative content	Marks	Part marks and guidance		
	b	$\text{DR } \frac{dr}{d\theta} = \frac{1}{2} \cos \theta (\sin \theta)^{-\frac{1}{2}} e^{\frac{1}{3} \cos \theta} +$ $(\sin \theta)^{\frac{1}{2}} \left(-\frac{1}{3} \sin \theta \right) e^{\frac{1}{3} \cos \theta}$ $\frac{dr}{d\theta} = \frac{1}{6} (\sin \theta)^{-\frac{1}{2}} e^{\frac{1}{3} \cos \theta} (3 \cos \theta - 2 \sin^2 \theta)$ $\frac{dr}{d\theta} = 0 \Rightarrow 3 \cos \theta - 2 \sin^2 \theta = 0$ $2 \cos^2 \theta + 3 \cos \theta - 2 = 0$ $\cos \theta = \frac{1}{2}, -2$ $\cos \theta \neq -2$ $\Rightarrow \sin \theta = \frac{\sqrt{3}}{2} \Rightarrow r = \sqrt{\frac{\sqrt{3}}{2} e^{\frac{1}{3} \cdot \frac{1}{2}}} = \sqrt{\frac{\sqrt{3}}{2} e^{\frac{1}{6}}}$	*M1(AO3.1a)A1(AO1.1)dep *M1(AO2.2a)M1(AO2.1)*A1(AO1.1)dep *A1(AO2.3)A1(AO2.2a)[7]	Attempt to differentiate using product at and chain rules. Setting r' to zero and factorising / cancelling to produce a quadratic equation in $\cos$ and / or $\sin$. Use of $\cos^2 + \sin^2 = 1$ to find 3 term quadratic equation in $\cos \theta$. Solving quadratic correctly Explicitly rejecting root AG. At least one intermediate step must be seen. <u>Examiner's Comments</u> Some candidates did not realise that they needed to find $\frac{dr}{d\theta}$. Others made errors when finding it, even when writing out the	Must be in the form $uv' + u'v$ with most one of u, v, u' or v' incorrect or omitted Or could be in $\sin^2 \theta$; $4 \sin^4 \theta + 9 \sin^2 \theta - 9 = 0$ $\sin^2 \theta = \frac{3}{4}, -3$ Rejects $\sin^2 \theta = -3$ and $\sin \theta = -\frac{\sqrt{3}}{2}$ Can be awarded even if rejection of root(s) was implicit.	

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Question			Answer/Indicative content	Marks	Part marks and guidance	
					product rule term-by-term. Where candidates progressed past these initial issues, they generally produced a coherent mathematical argument. Exemplar 4 	
					Centres should stress to candidates that an early error in a major problem can lead to most marks becoming inaccessible. Here, the candidate omits one key term ($e^{\frac{1}{2} \cos \theta}$) while differentiating a product. The resultant expression becomes impossible to meaningfully factorise. Consequently, the candidate cannot solve $\frac{dr}{d\theta} = 0$, meaning that they gain only 1 of the 7 marks available.	
			Total	11		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
3	a		$\ln(1 + \sin\theta) = 0 \Rightarrow 1 + \sin\theta = 1 \Rightarrow \sin\theta = 0$ so $\alpha = 0$ and $\beta = \pi$	M1 (AO1.1a) A1 (AO2.2a) [2]		
	b		$A = \frac{1}{2} \int_0^{\pi} (\ln(1 + \sin\theta))^2 d\theta$ $= 0.4162$ (4 sf) cao	M1 (AO1.2) A1 (AO1.1) [2]	Correct formula for area with r correctly substituted and their limits. Must be unambiguous but can be implied by correct answer / later work BC	Incorrect formula = M0A0 Condone missing $d\theta$



Question			Answer/Indicative content	Marks	Part marks and guidance	
	c		$\theta = \frac{\pi}{2} \Rightarrow r = \ln 2 = 0.6931$ (4 sf) which would be the diameter, D , of the circle But $A = 0.4162$ (4 sf) $\Rightarrow D$ $= 0.7280$ (4 sf) or $R =$ 0.3640 (4 sf) so the curve is not circular	M1 (AO3.1a) A1 (AO3.2a) [2]	or radius R $= 0.3466$ (4 sf) condone correct R or D without reasoning It must be clear that the r value would be the diameter of the circle; the calculation alone is insufficient for M1. M1 can be implied by area given $\pi \left(\frac{\ln 2}{2} \right)^2$ as or $R =$ 0.3466 (4 sf) (or $D =$ 0.6931) $\Rightarrow$ $A = 0.3773$ (4 sf) which is not 0.4162 (4 sf) Explanation must include comparison of R 's, D 's or A 's and conclusion. Allow correct working to 3 sf.	
			Total	6	<u>Examiner's Comments</u> Some poor notation was observed in (b) with the omission of the necessary outer brackets of the integral commonly seen. In (c) many candidates did not realise that the ordinate r did not equate to the radius of any potential circle.	

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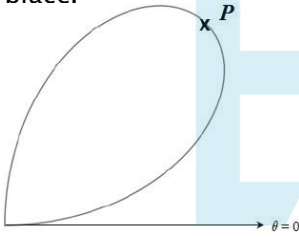


Question			Answer/Indicative content	Marks	Part marks and guidance	
4	a		$x = r \cos \theta, y = r \sin \theta \Rightarrow (r \cos \theta)^3 + (r \sin \theta)^3 = 2r \cos \theta \cdot r \sin \theta$ $\Rightarrow r (\cos^3 \theta + \sin^3 \theta) = 2 \cos \theta \sin \theta$ $\Rightarrow r = \frac{2 \cos \theta \sin \theta}{\cos^3 \theta + \sin^3 \theta}$	M1 (AO3.1a) A1 (AO1.1) [2]	Substitution May see "or $r = 0$ " but not required.	
	b		$f\left(\frac{1}{2}\pi - \theta\right) = \frac{2 \cos\left(\frac{1}{2}\pi - \theta\right) \sin\left(\frac{1}{2}\pi - \theta\right)}{\cos^3\left(\frac{1}{2}\pi - \theta\right) + \sin^3\left(\frac{1}{2}\pi - \theta\right)}$ $= \frac{2 \sin \theta \cos \theta}{\sin^3 \theta + \cos^3 \theta}$	M1 (AO1.1a) A1 (AO1.1) [2]	Correct substitution into <i>their</i> $f(\theta)$	
	c		So the line of symmetry is $\theta = \frac{\pi}{4}$	B1 (AO2.2a) [1]	Allow $y = x$. Must have $\theta =$	
	d		$r = f\left(\frac{1}{4}\pi\right) = \sqrt{2}$	B1 (AO1.1) [1]	BC	



Question			Answer/Indicative content	Marks	Part marks and guidance	
	e		$r = 0$ when $\theta = 0$. $r = 0$ also $\theta = \frac{\pi}{2}$ when In the range $0 < \theta < \frac{\pi}{2}$, $r > 0$ and continuous So there is a loop	B1 (AO3.1a) B1 (AO2.4) [2]	For both, ignore extras. Conclusion – both statements for r need to be mentioned <u>Examiner's Comments</u> Part (a) requiring the conversion of a cartesian equation to a polar equation was usually well done. The response to part(b) was identical to part (a) giving the hint for part (c). Part (e) had two parts. The first mark was for showing that when $r = 0$, θ took the values 0 and $\frac{\pi}{2}$. However, in order for there to be a loop, r had to be positive for all values in the range and this was often missed.	
			Total	8		



Question			Answer/Indicative content	Marks	Part marks and guidance	
5		a	(i) Maximum value of $\sin 2\theta = 1$ when $2\theta = \frac{\pi}{2} \Rightarrow \theta = \frac{\pi}{4}$ AG (ii) $r = 3$ (iii) P marked in correct place. 	M1 (AO 3.1a) A1 (AO 2.1) [2] B1 (A.O 1.1) [1] B1 (AO 1.1) [1]	Alternatively Differentiate r and set equal to 0 $r = 3 \sin 2\theta \Rightarrow \frac{dr}{d\theta} = 6 \cos 2\theta$ $= 0 \text{ when } \cos 2\theta = 0 \Rightarrow 2\theta = \frac{\pi}{2} \Rightarrow \theta = \frac{\pi}{4}$ $\Rightarrow r = 3 \sin \frac{\pi}{2} = 3$	
		b	D R $A = \frac{1}{2} \int_0^{\frac{\pi}{2}} r^2 d\theta = \frac{1}{2} \int_0^{\frac{\pi}{2}} (3 \sin 2\theta)^2 d\theta = \frac{9}{2} \int_0^{\frac{\pi}{2}} \sin^2 2\theta d\theta$ $= \frac{9}{4} \int_0^{\frac{\pi}{2}} (1 - \cos 4\theta) d\theta = \frac{9}{4} \left[\theta - \frac{1}{4} \sin 4\theta \right]_0^{\frac{\pi}{2}}$ $= \frac{9}{4} \left(\frac{\pi}{2} - \frac{1}{4} \sin 2\pi - 0 + \frac{1}{4} \sin 0 \right) = \frac{9\pi}{8}$	M1 (AO 3.1a) M1 (AO 3.1a) A1 (AO 1.1) M1 (AO 1.1) A1 (AO 1.1) [5]	Use of correct formula, ignore limits Use of double angle formula Integration Substitution of <i>their</i> limits into <i>their</i> integrated function	
			Total	9		



Question			Answer/Indicative content	Marks	Part marks and guidance	
6		a	$A = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$	B1 (AO 1.1) [1]		
		b	Stretch scale factor 1/3 parallel to x-axis $A^{-1} = \begin{pmatrix} \frac{1}{3} & 0 \\ 0 & 1 \end{pmatrix}$	M1 (AO 1.1) A1 (AO 2.2a) [2]	Must be complete description (except no need to specify 2-D)	
		c	Reflection in the line $y = -x$	B1 (AO 1.2) [1]		
		d	$BA = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} = \dots$ $\dots = \begin{pmatrix} 0 & -1 \\ -3 & 0 \end{pmatrix}$	M1 (AO 1.1a) A1 (AO 1.1) [2]	For understanding that the matrix representing successive transformations is the product in the correct order. ie BA, not AB	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		e	$(\mathbf{BA})^{-1} = -\frac{1}{3} \begin{pmatrix} 0 & 1 \\ 3 & 0 \end{pmatrix}$ $\mathbf{A}^{-1}\mathbf{B}^{-1} = \begin{pmatrix} \frac{1}{3} & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -\frac{1}{3} \\ -1 & 0 \end{pmatrix}$ $= -\frac{1}{3} \begin{pmatrix} 0 & 1 \\ 3 & 0 \end{pmatrix} = (\mathbf{BA})^{-1} \quad \text{AG}$	M1 (AO 1.1a) A1 (AO 1.1) [2]	For carrying out the procedure for inverting the matrix found in (d) (or BA worked out from scratch) OR M1 find $\mathbf{A}^{-1}\mathbf{B}^{-1}$ from scratch A1 demonstrate that $(\mathbf{BA})(\mathbf{A}^{-1}\mathbf{B}^{-1})$ is equal to $\mathbf{I}$ Calculation and statement of equality. Needs to be in the same form, not just an assertion	
			Total	8		

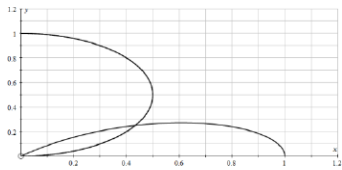


Question			Answer/Indicative content	Marks	Part marks and guidance		
7		a	Lower half of loop shaded	B1 (AO 1.1) [1]	Below axis of symmetry		
		b	$A = \frac{1}{2} \int_0^{\frac{1}{6}\pi} r^2 d\theta = 8 \int_0^{\frac{1}{6}\pi} \sin^2 3\theta d\theta$ $= 4 \int_0^{\frac{1}{6}\pi} (1 - \cos 6\theta) d\theta$ $= 4 \left[\theta - \frac{\sin 6\theta}{6} \right]_0^{\frac{1}{6}\pi}$ $= \frac{2}{3} \pi$	M1 (AO3.1a) M1 (AO1.1) A1 (AO1.1) A1 (AO1.1) [4]	Use of correct formula Use trig identity and integrate		
			Total	5			



Question			Answer/Indicative content	Marks	Part marks and guidance	
8		a	(i) $\sec \theta = \frac{r}{x}$ $\Rightarrow r = 2 - \frac{r}{x}$ $\Rightarrow r = \frac{2x}{x+1}$ $y^2 = r^2 - x^2 \Rightarrow y^2 = \left(\frac{2x}{x+1}\right)^2 - x^2$ (ii) Because the curve is y^2 =..... and so $x = p$ gives $y = \pm q$ (iii $a = -1$)	B1 (AO1.1) B1 (AO3.1a) B1 (AO3.1a) A1 (AO2.1) [4] B1 (AO2.4) [1] B1 (AO2.2a) [1]	AG	
		b	$V = \pi \int y^2 dx$ $= \pi \int \left(\left(\frac{2x}{x+1} \right)^2 - x^2 \right) dx = \pi \int \left(4 - x^2 - 4 \left(\frac{2x+2}{x^2+2x+1} \right) + \frac{4}{(x+1)^2} \right) dx$ $\pi \left[4x - \frac{x^3}{3} - 4 \ln(x+1)^2 - \frac{4}{(x+1)} \right]$ Limits are 0 and 1 $\Rightarrow V = \pi \left(\left(4 - \frac{1}{3} - 4 \ln 4 - 2 \right) - (0 - 0 - 0 - 4) \right)$ $= \pi \left(\frac{17}{3} - 4 \ln 4 \right)$	M1 (AO1.1a) M1 (AO3.1a) A1 (AO1.1) M1 (AO3.1a) A1 (AO1.1) B1 (AO3.1a) M1 (AO1.1) A1 (AO1.1) [8]	Use of formula any equivalent form that can be integrated Attempt to integrate $4 \ln 4$ and $8 \ln 2$ are equivalent	
			Total	14		



Question			Answer/Indicative content	Marks	Part marks and guidance	
9		i		B1 B1 B1 DB1	first curve a pproximatel y correct second curve approx correct intersection s on “axes” curves intersect ap proximately in the right place, dep on 1st two B marks	Ignore any extra parts of curves
				[4]	Examiner's Comments The sketches were often not neatly produced and the intercepts with the boundaries where $\theta = 0$ and $\theta = \frac{1}{2}\pi$ were rarely marked. The polar coordinates of the intersection were often correct; candidates who had drawn the part of $r = \cos 2\theta$ with negative r sometimes also gave $(1, \frac{1}{2}\pi)$. Sections of the graphs for which $r < 0$ were not penalised at this stage, but inevitably caused confusion in later parts. The determination of the area was sometimes correct, but many candidates integrated the part involving $r = \cos 2\theta$ with an upper limit of $\frac{1}{2}\pi$ instead of $\frac{1}{4}\pi$.	



Question			Answer/Indicative content	Marks	Part marks and guidance		
		ii	$r = 0.5$ $\theta = \frac{\pi}{6}$	B1 B1 [2]	Accept $\left(0.5, \frac{\pi}{6}\right)$ or $\left(\frac{\pi}{6}, 0.5\right)$ Examiner's Comments Diagrams to show what area was being calculated helped candidates to keep track of their working.		
		iii	Limit of $\frac{\pi}{6}$ For first: $A = \frac{1}{2} \int_0^{\pi/6} r^2 d\theta = \frac{1}{2} \int_0^{\pi/6} \sin^2 \theta d\theta = \frac{1}{4} \int_0^{\pi/6} (1 - \cos 2\theta) d\theta$ $= \frac{1}{4} \left(\theta - \frac{1}{2} \sin 2\theta \right) \quad \left(= \frac{1}{4} \left(\frac{\pi}{6} - \frac{\sqrt{3}}{4} \right) \right)$ For second: $A = \frac{1}{2} \int_{\pi/6}^{\pi/4} r^2 d\theta = \frac{1}{2} \int_{\pi/6}^{\pi/4} \cos^2 2\theta d\theta = \frac{1}{4} \int_{\pi/6}^{\pi/4} (1 + \cos 4\theta) d\theta$ $= \frac{1}{4} \left(\theta + \frac{1}{4} \sin 4\theta \right) \quad \left(= \frac{\pi}{48} + \frac{\sqrt{3}}{32} \right)$ Total area = $\frac{\pi}{16} - \frac{3\sqrt{3}}{32}$	B1 M1 DM1 A1 DM1 A1 M1 A1 [8]	Seen anywhere in either integral Formula for area applied to at least one curve ignoring limits Deal with $\sin^2 \theta$ correctly answer before limits deal with $\cos^2 2\theta$ correctly answer before limits Add their numerical areas from different limits (or subtract ap	$\frac{1}{2} \int (r_1^2 - r_2^2) d\theta$ could earn the first 6 marks	



Question			Answer/Indicative content	Marks	Part marks and guidance		
					appropriately) cao Examiner's Comments Candidates producing two areas based on the same limits had clearly not understood the underlying techniques and it was not unreasonable that this was reflected in the marks awarded for this part of the question.		
			Total	14			
10			DR $r = 0 \Rightarrow k = \frac{1}{8}$ $A = \frac{1}{2} \int_{-\frac{1}{8}\pi}^{\frac{1}{8}\pi} r^2 d\theta = \frac{1}{2} \int_{-\frac{1}{8}\pi}^{\frac{1}{8}\pi} (2 \cos 4\theta)^2 d\theta$ $= 2 \int_{-\frac{1}{8}\pi}^{\frac{1}{8}\pi} \cos^2 4\theta d\theta$ $= \int_{-\frac{1}{8}\pi}^{\frac{1}{8}\pi} (1 + \cos 8\theta) d\theta = \left[\theta + \frac{\sin 8\theta}{8} \right]_{-\frac{1}{8}\pi}^{\frac{1}{8}\pi}$ $= \left(\frac{1}{8}\pi + \frac{\sin \pi}{8} \right) - \left(-\frac{1}{8}\pi + \frac{\sin(-\pi)}{8} \right) = \frac{1}{4}\pi$	B1(AO 2.2a) M1(AO3. 1a) A1(AO1. 1) A1(AO1. 1) M1(AO3. 1a) A1(AO1. 1) [6]	Limits not required Correct form to be integrated Correct indefinite integral Using correct limits	Must be seen Must be seen A0 for decimal answer	
			Total	6	Must show		

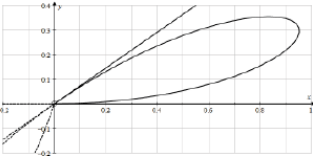
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Question			Answer/Indicative content	Marks	Part marks and guidance	
11		a	e.g. $r^4 \cos^4 \theta + r^4 \sin^4 \theta = r^2$ $r^2 = \frac{1}{\cos^4 \theta + \sin^4 \theta}$ $= \frac{1}{(\cos^2 \theta + \sin^2 \theta)^2 - 2 \sin^2 \theta \cos^2 \theta}$ $= \frac{1}{1 - \frac{1}{2} \sin^2 2\theta} = \frac{2}{2 - \sin^2 2\theta}$	B1(AO1.1) B1(AO1.1) M1(AO2.1) A1(AO2.2a) [4]	Substitute $x = r \cos \theta$, $y = r \sin \theta$ Use $x^2 + y^2 = r^2$ Rearrange and attempt to use relevant trigonometric identity AG	
		b	Maximum value of r occurs when $\sin 2\theta = 1$ $r^2 = \frac{2}{2-1} = 2 \Rightarrow r = \sqrt{2}$	M1(AO1.1) A1(AO2.2a) [2]	AG	
			Total	6		

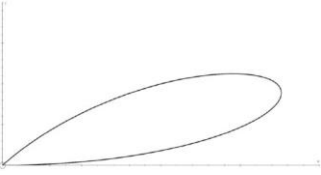


Question			Answer/Indicative content	Marks	Part marks and guidance	
12		i		B1	Single enclosed loop in	Examiner's Comments The majority of candidates scored full marks. The most common mistake was to place the loop too high in the quadrant. Stronger candidates were clearly making an effort to make $\theta = 0$ a proper looking tangent whereas others seemed to have a shaky concept of what that meant.
		i		B1	1st quadrant tangent at origin (not necessarily seen) less	
		i		B1	than $\theta = \frac{\pi}{4}$	
		i		B1	$\theta = \frac{\pi}{5}$ stated	
		i			$\theta = 0$ stated	
		i			-1 any extra "tangents" if 2 are correct	
Copyright © 2024 Exam Papers Practice			ii $\theta = \frac{\pi}{10}$	B1	For θ . Allow cartesian $y = \left(\tan \frac{\pi}{10}\right)x$	
		ii	$x = r \cos \theta = \cos \frac{\pi}{10} (\approx 0.951)$	B1		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
13		i		B1	Enclosed loop in first quadrant with origin as pole	
		ii		B1	Looking symmetric with line of symmetry around $\theta = \frac{\pi}{6}$ Take one off full marks for more loops Examiner's Comments Most attempted to draw an enclosed loop of some description. Very few drew loops in other quadrants. Those who had found tangents at the pole and marked them on graph produced better diagrams and avoided making $\theta = \frac{\pi}{2}$ look like a tangent.	
		ii	Area = $\frac{1}{2} \int_0^{\pi/3} r^2 d\theta = 2 \int_0^{\pi/3} \sin^2 3\theta d\theta$	M1	Correct formula plus limits	Must include $\frac{1}{2}$
		ii	$= \int_0^{\pi/3} (1 - \cos 6\theta) d\theta = \left[\theta - \frac{1}{6} \sin 6\theta \right]_0^{\pi/3}$	M1	For obtaining fn in form to integrate using double angle formulae	
		ii	$= \frac{\pi}{3}$	A1	Integral Ft lack of $\frac{1}{2}$	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii		A1	Answer www	
					Examiner's Comments	
					This part was largely well-done. Errors included the failure to find an expression for $\sin^2 3\theta$ which could be integrated.	
		ii	Alternative: Starting from given equation: Eliminating x and y	M1		
		ii	Get r	M1		
		iii	$\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$	M1	Obtaining $\sin 3\theta$ as a function of θ	
		iii	$y = r \sin \theta \Rightarrow \sin \theta = \frac{y}{r}$ and $r^2 = x^2 + y^2$	A1	A correct expression	Examiner's Comments This last part was often done quite well, though a surprising number could not find an expression for $\sin 3\theta$. Once again there was a fair amount of fudging where the correct, but given, answer appeared at the end.
		iii	$r = 2 \sin 3\theta = 6 \sin \theta - 8 \sin^3 \theta = \frac{6y}{r} - \frac{8y^3}{r^3}$	M1	Eliminate θ	
		iii	$\Rightarrow r^4 = 6y r^2 - 8y^3$	M1	Eliminate r	
		iii	$\Rightarrow (x^2 + y^2)^2 = 6(x^2 + y^2)y - 8y^3 = 6x^2 y - 2y^3$	A1	ag	
					Examiner's Comments	
					This last part was often done quite well, though a surprising number could not find an expression for $\sin 3\theta$. Once again there was a fair amount of fudging where the correct, but given, answer appeared at the end.	
		iii	Alternative: Starting from given equation: Eliminating x and y	M1		
		iii	Get r	M1		
		iii	$r = 6 \sin \theta - 8 \sin^3 \theta$	A1		
		iii	Obtain triple angle formula	M1		
		iii	Ans	A1		



Question			Answer/Indicative content	Marks	Part marks and guidance
			Total	11	



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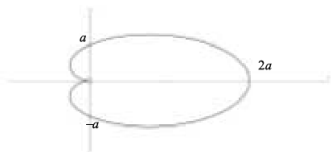
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Question			Answer/Indicative content	Marks	Part marks and guidance	
14		i		B1 B1 B1	Enclosed loop with axes tangential $\theta = \frac{\pi}{4}$ is a line of symmetry drawn and named	Ignore anything in other quadrants
		i	P is at $r = 5, \theta = \frac{\pi}{4}$	B1	For both <u>Examiner's Comments</u> A number failed to note the precise demands of the question and drew curves for $\theta > \frac{1}{2}\pi$, or failed to state the equation of the line of symmetry or the coordinates of P. The sketch was also expected to show that candidates understood that $\theta = 0$ and $\theta = \frac{1}{2}\pi$ were tangents to the curve at the pole.	
		ii	$\text{Area} = \frac{1}{2} \int_0^{\pi/2} r^2 d\theta = \frac{1}{2} \int_0^{\pi/2} 25 \sin^2 2\theta d\theta$ $= \frac{25}{4} \int_0^{\pi/2} (1 - \cos 4\theta) d\theta = \frac{25}{4} \left[\theta - \frac{1}{4} \sin 4\theta \right]_0^{\pi/2}$ $= \frac{25}{4} \left(\left(\frac{\pi}{2} - 0 \right) - (0) \right) = \frac{25\pi}{8}$	M1 M1 A1	Correct formula with r substituted. Correct method of integration including limits www Examiner's Comments Most candidates scored well here, adopting the correct way to carry out the integration.	
		iii	Equation is of the form $x + y = c$	B1		



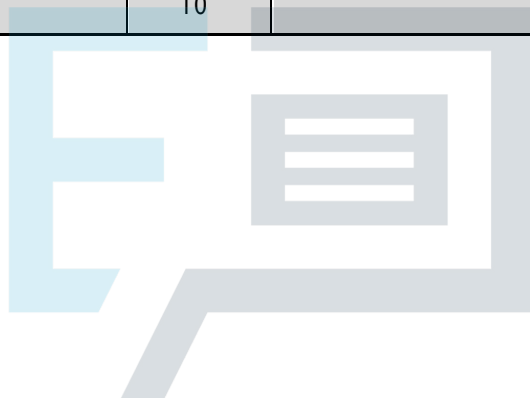
Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$P \text{ is } \left(\frac{5}{\sqrt{2}}, \frac{5}{\sqrt{2}} \right) \text{ oe}$ $\Rightarrow x + y = 5\sqrt{2}$	B1 B1	Ft. $x + y = c$ where c comes from their P. <u>Examiner's Comments</u> This part defeated many candidates. Because P was the point furthest from O , some candidates thought <u>dr</u> that <u>dθ</u> had to be involved and even set equal to zero.	
		iv	$r = 5\sin 2\theta = 10\sin\theta\cos\theta$	M1	Square and convert r^2	
		iv	$\Rightarrow r^2 = 100\sin^2\theta\cos^2\theta = 100\left(\frac{y}{r}\right)^2\left(\frac{x}{r}\right)^2$	M1	Substitute for r and θ	
		iv	$\Rightarrow (x^2 + y^2)^3 = 100x^2y^2$	A1	NB Answer given	
			Total	13		



Question			Answer/Indicative content	Marks	Part marks and guidance	
15		i	$a(1 + \cos\theta) = 0$ when $\cos\theta = -1$	M1	soi	
		i	$\Rightarrow \theta = \pi$	A1	Only this answer: A0 if anything else Examiner's Comments Candidates should label the axes of the polar graph and mark any intersections with the axes.	
		ii		B1	Correct shape, correct orientation, roughly symmetric	
		ii		B1	All 3 intersections on axes indicated, cusp at pole dep on 1st B. Examiner's Comments Candidates should label the axes of the polar graph and mark any intersections with the axes.	
		iii	$r = a(1 + \cos\theta)$	M1	Use of formula with limits	
		iii	$A = \frac{1}{2} \int_0^{2\pi} r^2 d\theta = \frac{1}{2} \int_0^{2\pi} a^2 (1 + \cos\theta)^2 d\theta$			
		iii			Condone omission of a^2	
		iii	$= \frac{a^2}{2} \int_0^{2\pi} (1 + 2\cos\theta + \cos^2\theta) d\theta$	A1		
		iii			Dealing with $\cos^2$	
		iii	$= \frac{a^2}{2} \int_0^{2\pi} (1 + 2\cos\theta + \frac{1}{2}(\cos 2\theta + 1)) d\theta$	M1		
		iii			Condone omission of a^2	
		iii		A1		



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$= \frac{a^2}{2} \left[\theta + 2 \sin \theta + \frac{1}{2} \left(\frac{1}{2} \sin 2\theta + \theta \right) \right]_0^{2\pi} = \frac{a^2}{2} \left(2\pi + 0 + \frac{1}{2} (0 + 2\pi) \right)$	M1	Substitute limits dep on 2nd M	
					Examiner's Comments	
					Was often done well, although sometimes the a^2 disappeared part way through (or worse, the dimension of each term changed) and a few candidates could not deal with integrating $\cos \theta$.	
		iii	$= \frac{a^2}{2} (3\pi) = \frac{3\pi a^2}{2}$	A1		
			Total	10		



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Question			Answer/Indicative content	Marks	Part marks and guidance	
16		i	Substitute $r^2 = x^2 + y^2, x = r \cos \theta$	M1		
		i		A1		
		i	$\Rightarrow r^2 - r \cos \theta = r \Rightarrow r = 1 + \cos \theta$	A1	cao Examiner's Comments This was mostly answered correctly. The most common error was to state that $x = r \sin \theta$. Candidates who drew a quick sketch to prompt them did not make this error.	
		ii		B1	Cardioid (General shape)	
		ii		B1	Correct shape at pole, $r = 2$ and symmetric Examiner's Comments Those candidates who made brief supporting comments such as "a cardioid" etc enabled the examiner to award full marks when their sketch was not fully convincing. The second mark was for the accuracy of the curve at the pole. A number of sketches did not include the pole. Some candidates who did include the pole did not draw their graph in a way that made the behaviour at the pole clear.	e.g. cusp clearly at pole, vertical tangent at $r = 2$
		iii	Line cuts curve at (0, 1) and (2, 0) Total area = $2 \times \frac{1}{2} \times \int_0^\pi (1 + \cos \theta)^2 d\theta$	B1		
		iii	$= \int_0^\pi (1 + 2\cos \theta + \cos^2 \theta) d\theta = \int_0^\pi \left(1 + 2\cos \theta + \frac{1 + \cos 2\theta}{2} \right) d\theta$	M1	Formula for area used	Sight of expansion and attempt to integrate



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$= \left[\frac{3}{2}\theta + 2\sin\theta + \frac{1}{4}\sin 2\theta \right]_0^{\pi} = \frac{3}{2}\pi$ $\text{area in 1st quadrant} = \frac{1}{2} \times \int_0^{\frac{1}{2}\pi} (1 + \cos\theta)^2 d\theta$	A1		
		iii	$= \frac{1}{2} \left[\frac{3}{2}\theta + 2\sin\theta + \frac{1}{4}\sin 2\theta \right]_0^{\frac{1}{2}\pi} = \frac{3}{8}\pi + 1$	A1		
		iii	Area under line in 1st quadrant = 1 $\Rightarrow \text{Area enclosed by line and curve} = \frac{3}{8}\pi + 1 - 1 = \frac{3}{8}\pi$	M1		
		iii	$\Rightarrow \text{ratio} = \left(\frac{3}{2}\pi - \frac{3}{8}\pi \right) : \frac{3}{8}\pi = 3:1$	A1	Or ratio 1: 3	
					Examiner's Comments	
					This was a challenging question for all but the highest scoring candidates.	
					For six marks, candidates should be aware that explicit working needs to be seen. Some candidates appeared to be using a calculator to complete the integration. Those candidates who made a sketch were able to refer to this in their working and made the most progress. Students should be encouraged to provide a sketch even if the question does not ask for it explicitly.	
			Total	11		

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